The Eye of Agamotto

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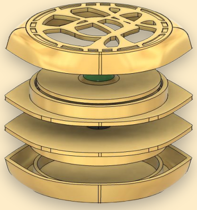
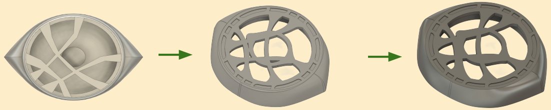
Journey

Inspiration: The Eye of Agamotto, a magical amulet used by Doctor Strange



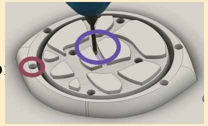
Design Goals:
- Oblong, eye shape with smooth edges
- Fine engravings and overhanging branches
- Green thermoformed "Time Stone"

Design for Manufacturing

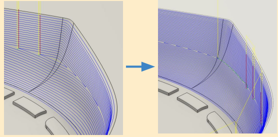


9/17: *Initial* 10/12 10/21: *Final*

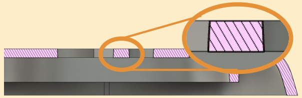
Challenge #1
- Fine details limited by end mill diameter (1/16") & need surface area for ejector pins ➜ widen channels



Challenge #2
- Edges/many surfaces may lead to part getting stuck in mold ➜ smoother surface finish, draft angles of 2°

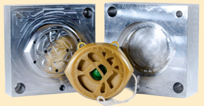


Rough Smooth



Manufacturing

Injection Molding: used the Arburg Injection Molder to produce 50 top rings, 50 bottom rings, and 100 body parts using 3 aluminum molds



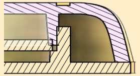
Thermoform: used thermoformer, punch, & spray paint to produce 50 parts with SLA printed mold

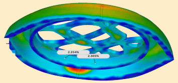
Difficulties

- **Most Difficult:** Machining detailed ring molds
  - Tool breakage, rough surface ➜ decreased material removal rate
  - Unable to machine narrow groove ➜ widen channels
- **Additional Difficulties**
  - **Bottom Ring:** Oversized pressfit ➜ scale for shrinkage
  - **Body:** IM burning ➜ lowered velocity to 0.5 m/s and pressure to 400 psi
  - **Thermoform Part:** Difficult to center for die-cutter ➜ sharp "cliff" edge in mold for alignment

Press Fit Analysis

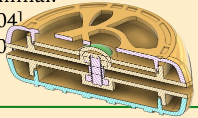
Press Fit Design: Used a commonly accepted value of 0.015" and ensured correct machined dimensions via shrinkage simulations and analysis

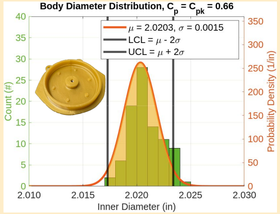


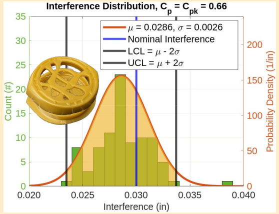


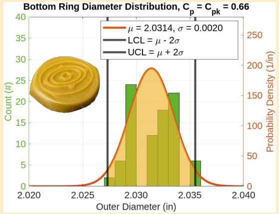
Press Fit Outcomes: Our measurements differed from our intended specification limits (SL). However, since both parts had the same shrinkage, the mean interference was close to nominal.

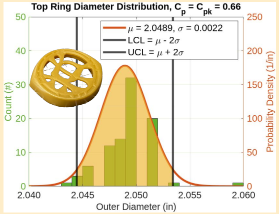
- **Body SL:** [2.02, 2.04]
- **Ring: SL:** [1.99, 2.0











Lessons Learned

- Account for machine downtime in manufacturing and know your cycle times
- Iteration is the learning process
- Consider manufacturability and ease of assembly from the start
- Thorough documentation helps everyone on the team